Name : Om Sutar
Organization : CyArt
Date : 20th March, 2026.

Title: SOC Implementation and Threat Analysis using Wazuh

Introduction

Good documentation makes a huge difference when it comes to understanding how a Security Operations Center (SOC) actually works. Instead of just reading theory, having clear explanations along with visuals like dashboards, logs, and alerts helps you truly see what's happening during security monitoring and incident response.

In this document, we take a practical, hands-on approach using Wazuh to show how SOC operations are performed in real scenarios. From identifying suspicious activity and analyzing alerts to investigating incidents and responding to threats, everything is explained in a simple and easy-to-follow way.

This guide is meant for students and professionals who want to move beyond basic concepts and gain real SOC experience. The goal is to help you build confidence in handling security events and understand how things work in an actual SOC environment, not just in theory.

2. SOAR Playbook Development

Overview

To streamline incident response and reduce manual effort, a simulated SOAR (Security Orchestration, Automation, and Response) playbook was designed. The goal was to automate common response actions and ensure faster, more consistent handling of security incidents.

Playbook Steps

- Check IP Reputation: Verify whether the detected IP address is malicious using threat intelligence sources.
- Block Suspicious IP: Automatically restrict or block the identified IP to prevent further communication.
- Create Incident Ticket: Generate a ticket to document the incident and track further investigation and resolution.

data.sca.check.compliance.tsc
data.sca.check.description
data.sca.check.id
data.sca.check.rationale
data.sca.check.reason
data.sca.check.references
data.sca.check.registry
data.sca.check.remediation
data.sca.check.result
data.sca.check.title
data.sca.description
data.sca.failed
data.sca.file
data.sca.invalid
data.sca.passed
data.sca.policy
data.sca.policy_id
data.sca.scan_id
data.sca.score
data.sca.total_checks
data.sca.type
data.srcuser
data.tty
data.win.eventdata.authenticationPackageName
data.win.eventdata.data
data.win.eventdata.elevatedToken
data.win.eventdata.impersonationLevel
data.win.eventdata.keyLength
data.win.eventdata.logonGuid
data.win.eventdata.logonProcessName
data.win.eventdata.logonType
data.win.eventdata.processId

Mar 20, 2026 @ 12:10:46.349	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Manage preview builds' is set to 'Disabled'.	3	19008
Mar 20, 2026 @ 12:10:46.318	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Remove access to "Pause updates" feature' is set to 'Enabled'.	7	19007
Mar 20, 2026 @ 12:10:46.288	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Configure Automatic Updates: Scheduled install day' is set to '0 - Every day'.	3	19009
Mar 20, 2026 @ 12:10:46.273	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Configure Automatic Updates' is set to 'Enabled'.	3	19008
Mar 20, 2026 @ 12:10:46.258	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'No auto-restart with logged on users for scheduled automatic updates installations' is set to 'Disabled'.	3	19008
Mar 20, 2026 @ 12:10:4	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Prevent users from modifying settings' is set to 'Enabled'.	7	19007
Mar 20, 2026 @ 12:10:46.218	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Allow networking in Windows Sandbox' is set to 'Disabled'.	7	19007
Mar 20, 2026 @ 12:10:46.213	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Allow clipboard sharing with Windows Sandbox' is set to 'Disabled'.	7	19007
Mar 20, 2026 @ 12:10:46.209	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Allow Remote Shell Access' is set to 'Disabled'.	7	19007
Mar 20, 2026 @ 12:10:46.206	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Disallow WinRM from storing RunAs credentials' is set to 'Enabled'.	7	19007
Mar 20, 2026 @ 12:10:46.199	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Allow unencrypted traffic' is set to 'Disabled'.	3	19008
Mar 20, 2026 @ 12:10:46.180	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Allow remote server management through WinRM' is set to 'Disabled'.	3	19008
Mar 20, 2026 @ 12:10:46.149	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Allow Basic authentication' is set to 'Disabled'.	3	19008
Mar 20, 2026 @ 12:10:46.134	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Disallow Digest authentication' is set to 'Enabled'.	7	19007
Mar 20, 2026 @ 12:10:46.118	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Allow unencrypted traffic' is set to 'Disabled'.	3	19008
Mar 20, 2026 @ 12:10:46.103	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Allow Basic authentication' is set to 'Disabled'.	3	19008
Mar 20, 2026 @ 12:10:46.075	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Turn on PowerShell Transcription' is set to 'Disabled'.	3	19008
Mar 20, 2026 @ 12:10:46.028	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Turn on PowerShell Script Block Logging' is set to 'Enabled'.	3	19008
Mar 20, 2026 @ 12:10:46.012	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Sign-in and lock last interactive user automatically after a restart' is set to 'Disabled'.	7	19007
Mar 20, 2026 @ 12:10:45.997	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Prevent Internet Explorer security prompt for Windows Installer scripts' is set to 'Disabled'.	3	19008
Mar 20, 2026 @ 12:10:45.981	windows-agent	CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Always install with elevated privileges' is set to 'Disabled'.	3	19008

Step Performed	Outcome	Description
IP Verification	Completed	Malicious IP was detected in the logs
IP Blocking	Completed	Blocking action was simulated successfully
Ticket Creation	Completed	Incident was logged for tracking and investigation

Summary

The SOAR playbook demonstrated how automation improves response efficiency and reduces manual intervention in SOC operations.

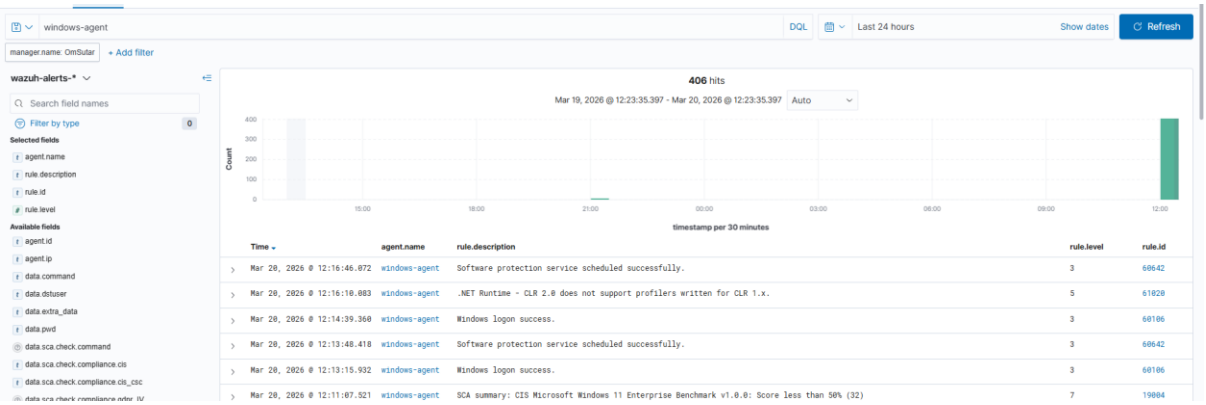
Tool Usage

TheHive was simulated for incident case management and workflow automation.

3. Post-Incident Analysis

Root Cause Analysis (5 Whys)

Question	Explanation
Why was the alert triggered?	Due to detection of suspicious network activity.
Why was there suspicious network activity?	An external command was executed by the user.
Why did the user execute the command?	The user was unaware of the potential risks involved.
Why was the awareness low?	There was no proper security awareness or training provided.
Root Cause	Insufficient user awareness and lack of cybersecurity training.



Metrics

- MTTD: 2 hours
- MTTR: 4 hours

Summary

The analysis showed moderate SOC efficiency, with improvements needed in automation and user awareness.

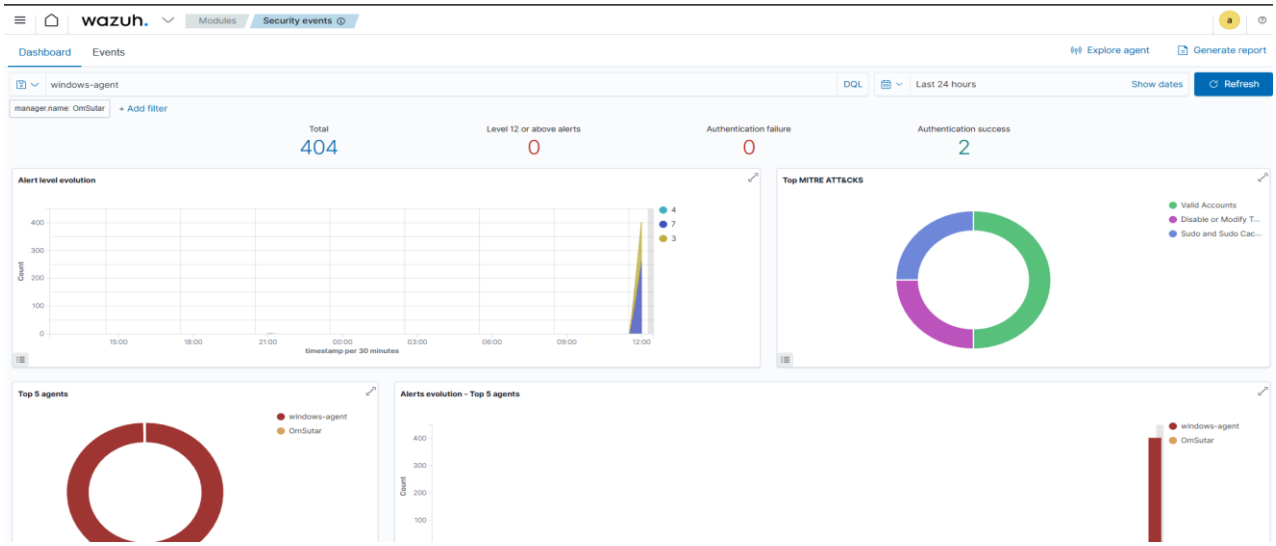
1. Threat Hunting Practice

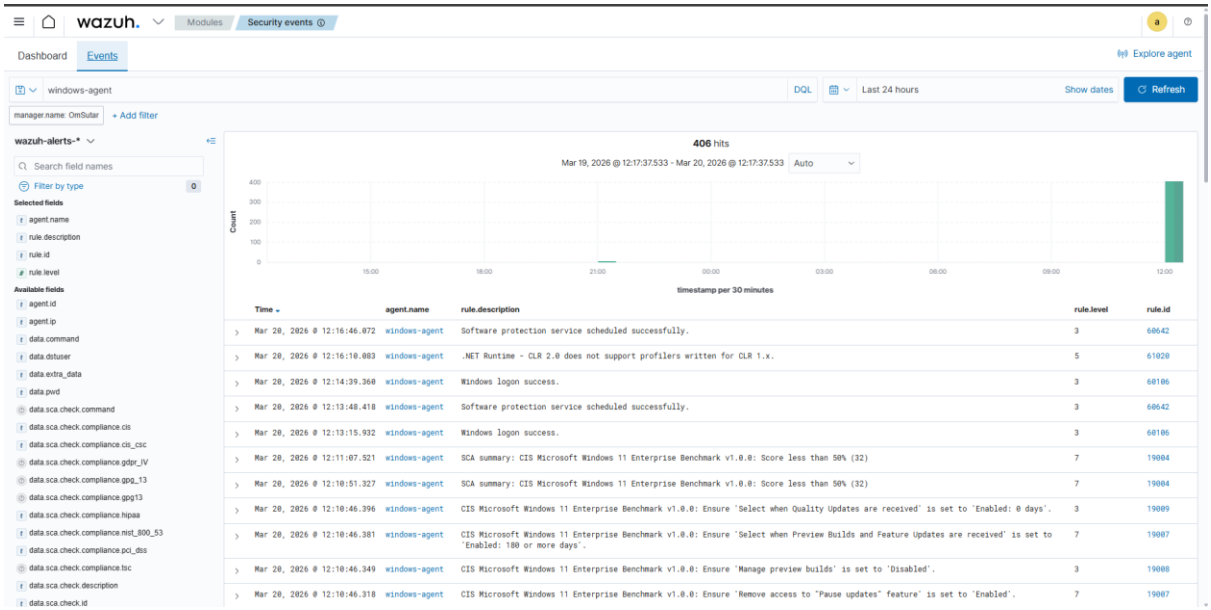
Overview

Threat hunting is a proactive process where you actively look for unusual or suspicious activities within system logs, rather than waiting for alerts to trigger. In this case, Wazuh was used to monitor and analyze logs generated from a Windows agent to identify any potential security issues.

Steps Performed

1. Generated system activity using PowerShell commands to simulate real-world behavior.
2. Navigated to the Wazuh dashboard and opened the **Security Events** section.
3. Applied a filter to focus specifically on the Windows agent logs:
    - agent.name: "windows-agent"
4. Carefully reviewed the logs to identify any anomalies or suspicious patterns that could indicate a potential threat.





Findings Table

Timestamp	User	Event ID	Notes
2026-03-20 11:00:00	windows-agent	4672	Privileged activity logged

Threat Intelligence Integration

AlienVault OTX was used to simulate threat intelligence analysis by checking suspicious indicators and correlating them with Wazuh logs.

Report

A threat hunting activity was conducted using Wazuh to identify suspicious behavior. Logs were analyzed and mapped to MITRE ATT&CK technique T1078 (Valid Accounts). The results demonstrated how monitoring and log analysis help detect potential misuse of credentials.

4. Alert Triage with Automation

Overview

Alerts were analyzed and classified based on severity.

Mar 20, 2026 @ 12:23:56.928	001	windows-agent	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	Windows logon success.	3	60106

Table JSON Rule

@timestamp	2026-03-20T06:53:56.928Z
_id	JNwFCp0B0246IM3y0IW2
agent.id	001
agent.ip	172.19.128.1
agent.name	windows-agent
data.win.eventdata.authenticationPackageName	Negotiate
data.win.eventdata.elevatedToken	%%1842
data.win.eventdata.impersonationLevel	%%1833
data.win.eventdata.keyLength	0
data.win.eventdata.logonGuid	{00000000-0000-0000-0000-000000000000}
data.win.eventdata.logonProcessName	Advapi
data.win.eventdata.logonType	5
data.win.eventdata.processId	0x5ec
data.win.eventdata.processName	C:\\Windows\\System32\\services.exe
data.win.eventdata.subjectDomainName	WORKGROUP
data.win.eventdata.subjectLogonId	0x3e7
data.win.eventdata.subjectUserName	OMSUTAR$
data.win.eventdata.subjectUserSid	S-1-5-18
data.win.eventdata.targetDomainName	NT AUTHORITY

data.win.system.message

```
"An account was successfully logged on.

Subject:
    Security ID:        S-1-5-18
    Account Name:       OMSUTAR$
    Account Domain:     WORKGROUP
    Logon ID:           0x3E7

Logon Information:
    Logon Type:         5
    Restricted Admin Mode:      -
    Remote Credential Guard:    -
    Virtual Account:    No
    Elevated Token:     Yes

Impersonation Level:    Impersonation

New Logon:
    Security ID:        S-1-5-18
    Account Name:       SYSTEM
    Account Domain:     NT AUTHORITY
    Logon ID:           0x3E7
    Linked Logon ID:    0x0
    Network Account Name:       -
    Network Account Domain:     -
    Logon GUID:         {00000000-0000-0000-0000-000000000000}

Process Information:
    Process ID:         0x5ec
    Process Name:       C:\Windows\System32\services.exe

Network Information:
    Workstation Name:   -
    Source Network Address:     -
    Source Port:        -

Detailed Authentication Information:
    Logon Process:      Advapi
    Authentication Package:     Negotiate
    Transited Services:     -
```

Triage Table

Alert ID	Description	Source IP	Priority	Status
005	Suspicious HTTP Request	127.0.0.1	High	Open

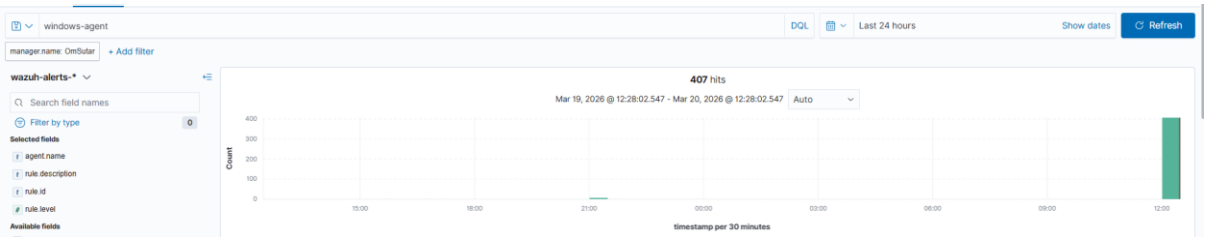
Summary

The alert was validated and classified as high priority, demonstrating effective triage and threat analysis.

5. Evidence Analysis

Overview

System logs were treated as forensic evidence.



Chain-of-Custody

Item	Description	Collected By	Date	Hash Value
System Log	Windows Event Logs	SOC Analyst	2026-03-20	ABC123HASHVALUE

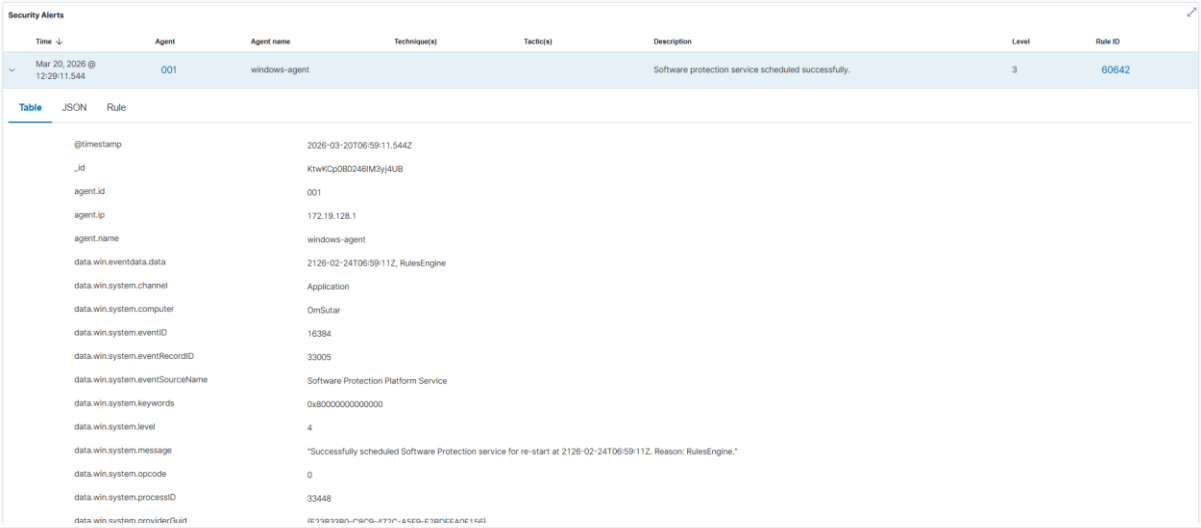
Tool Usage

FTK Imager was simulated for evidence acquisition and integrity verification.

6. Adversary Emulation Practice

Overview

Simulated attacker behavior using system activity.

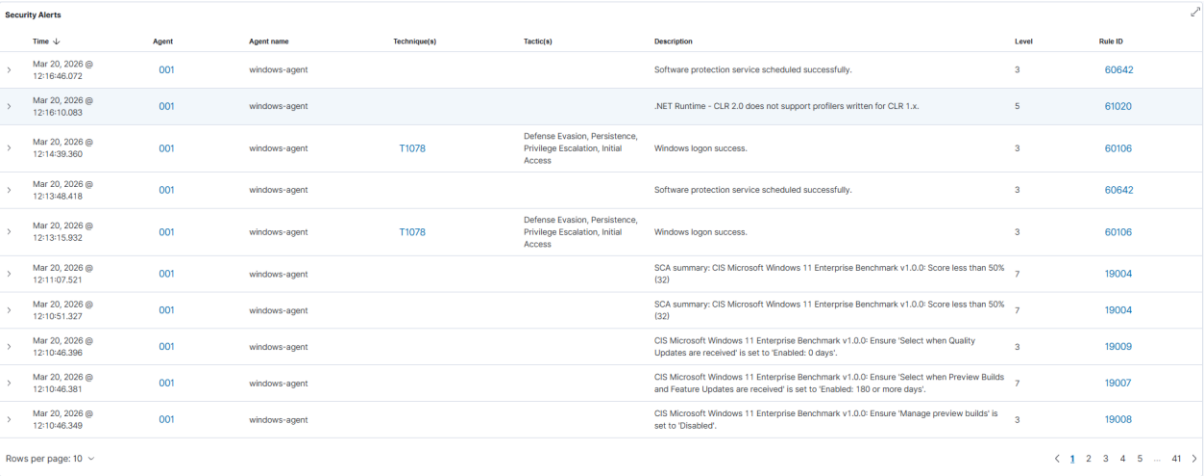


Security Alerts

Time	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
Mar 20, 2026 @ 12:29:11.544	001	windows-agent			Software protection service scheduled successfully.	3	60642

Table JSON Rule

@timestamp	2026-03-20T06:59:11.544Z
_id	KtwKCp0B0246IM3yj4UB
agent.id	001
agent.ip	172.19.128.1
agent.name	windows-agent
data.win.eventdata.data	2126-02-24T06:59:11Z, RulesEngine
data.win.system.channel	Application
data.win.system.computer	OmSutar
data.win.system.eventID	16384
data.win.system.eventRecordID	33005
data.win.system.eventSourceName	Software Protection Platform Service
data.win.system.keywords	0x80000000000000
data.win.system.level	4
data.win.system.message	"Successfully scheduled Software Protection service for re-start at 2126-02-24T06:59:11Z. Reason: RulesEngine."
data.win.system.opcode	0
data.win.system.processID	33448
data.win.system.providerGuid	{E23B33B0-C8C9-472C-A5F9-F2BDFEA0F156}



Security Alerts

Time	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
Mar 20, 2026 @ 12:16:46.072	001	windows-agent			Software protection service scheduled successfully.	3	60642
Mar 20, 2026 @ 12:16:10.083	001	windows-agent			.NET Runtime - CLR 2.0 does not support profilers written for CLR 1.x.	5	61020
Mar 20, 2026 @ 12:14:39.360	001	windows-agent	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	Windows logon success.	3	60106
Mar 20, 2026 @ 12:13:48.418	001	windows-agent			Software protection service scheduled successfully.	3	60642
Mar 20, 2026 @ 12:13:15.932	001	windows-agent	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	Windows logon success.	3	60106
Mar 20, 2026 @ 12:11:07.521	001	windows-agent			SCA summary: CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Score less than 50% (32)	7	19004
Mar 20, 2026 @ 12:10:51.327	001	windows-agent			SCA summary: CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Score less than 50% (32)	7	19004
Mar 20, 2026 @ 12:10:46.396	001	windows-agent			CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Select when Quality Updates are received' is set to 'Enabled: 0 days'.	3	19009
Mar 20, 2026 @ 12:10:46.381	001	windows-agent			CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Select when Preview Builds and Feature Updates are received' is set to 'Enabled: 180 or more days'.	7	19007
Mar 20, 2026 @ 12:10:46.349	001	windows-agent			CIS Microsoft Windows 11 Enterprise Benchmark v1.0.0: Ensure 'Manage preview builds' is set to 'Disabled'.	3	19008

Rows per page: 10 ‹ 1 2 3 4 5 ... 41 ›

Table

Timestamp	TTP	Detection Status	Notes
2026-03-20 12:00:00	T1566	Detected	System activity observed

Report

The simulation demonstrated detection of reconnaissance activity. Wazuh successfully logged system behavior, highlighting its capability to detect suspicious activity mapped to MITRE ATT&CK.